

Automated Data or Lead Verification System (ADLVS)

Requirement Document

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1 Executive Summary

The goal is to develop a robust, tech-forward application that automates the verification of sales leads and contractor data. By transitioning from manual workflows to a multi-source, AI-driven automation engine, We aims to achieve a significant reduction in processing time while maintaining peak data integrity using multi-source cross-referencing and AI-driven workflows.

1.1 Key Performance Indicators (KPIs)

- **Current State:** 24 hours (Manual)
- **Target State:** < 3 hours (Automated). This time will vary depending on the size of the data and external API limitations if any.
- **Human-in-the-loop:** Automation handles the bulk; humans only intervene on "Failed" or "Inconclusive" flags.

2 Functional Requirements

2.1 Multi-Source Verification Engine

The system will support multiple types of verification for each data point. Below table provides an indicative verification tools / sources, based on our best knowledge. These tools are sources will be made configurable as much as possible. While configuring every lead, respective user can define the tool / source to be used for data verification.

Data Point	Primary Tool/Source	Secondary / Fallback Source
Email Address	Inbuilt verification (Custom SMTP Check) / ZeroBounce	Hunter / TowerData / Neverbounce
Phone Number	LinkedIn / Sales Navigator	AI Voice Bot (Workflow Execution)
Job Title	LinkedIn Individual Profile	Company Website "About Us"
Company Size	LinkedIn Insights	Glassdoor / Bloomberg
Industry Type	LinkedIn Company Page	Bloomberg
Physical Address	Google Maps API	Official Company Website

Website	DNS Lookup	URL validation
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2.2 Email Address verification

A Custom SMTP (Simple Mail Transfer Protocol) Check is a deep-level verification technique used to confirm the existence of a mailbox without actually sending an email as a default mechanism. In addition to this default mechanism, system will provide option to integrate with external REST APIs to verify emails. Any external integrations can be configured via an admin interface and user can choose the appropriate verification methods, while creating the lead.

2.2.1 Technical Considerations for AVO

To ensure this mechanism remains "robust and tech-savvy," several engineering safeguards are implemented:

- **Proxy Rotation:** To prevent the AVO server's IP from being blacklisted by major providers (like Gmail or Outlook), the system rotates through multiple IP addresses during SMTP checks.
- **Parallel Execution:** These checks are triggered simultaneously with other API calls to ensure the verification time is optimized.
- **Timeout Management:** The system is configured with strict timeout limits to prevent "slow" mail servers from delaying the entire batch.

2.3 AI Voice Bot & Dialer Integration for Phone number verification

The system incorporates an AI-driven voice agent (via platforms like Vapi or Bland AI) to verify identity and connectivity.

- **The "Verification Ping":** The bot executes pre-defined, campaign-specific scripts to confirm if a lead is still reachable or employed at the listed firm.
- **Advanced Telephony:** Supports board numbers, direct lines, and mobile numbers.
- **Intelligent Analysis:** The bot generates call recordings and transcripts will be enabled for telemarketing. It automatically verifies transcript data for expected

"Yes/No" results. If the results are not matching the expected data, but call continued, system will mark these results for manual intervention.

- **Security & Compliance:** Strictly limited to identity verification; no OTP or password handling is permitted.

2.4 Other data points verification

Data points like Job title or any other data points will be verified via LinkedIn, company web site or any other industry standard professional social media sites. AI tool should compare the input data with online data and identify similarities like IT manager in data entry vs Infrastructure manager in online and mark them as matching. If the data points in the portal are not matching with online content, then a comparison record need to be created in DB and mark for manual intervention without declaiming the validation.

2.5 Exception Handling (Human-in-the-Loop)

The system calculates a Final Confidence Score (S_{total}) to drive autonomous decision-making. The system will assign a **Confidence Score** (0-100%).

- **Auto-Approved (Score > 85%):** Data is passed without human review.
- **Manual Queue (Score 50% - 84%):** Sent to a human-in-the-loop for a quick "sanity check".
- **Fail (Score < 50%):** Highlighted for user correction.

3 Technical Architecture

ADLVS is built on a **microservices architecture** to ensure scalability and robustness:

- **Backend:** Python (FastAPI/Django) for high-speed data processing.
- **Automation:** Selenium or Playwright for web scraping with proxy rotation to avoid LinkedIn/Glassdoor bans.
- **Database & Caching:** PostgreSQL for structured data and Redis to cache results and optimize API costs.
- **Parallel Processing:** A Queue Manager triggers simultaneous verifications, configured to respect target website rate limits.
- **Deployment Options:** Native support for Azure, AWS, or On-Premise infrastructure, optimized for a minimum of 100,000 leads per day.

4 Security & Compliance

4.1 Authentication & Authorization

- Implements industry-standard OAuth 2.0 for authentication and authorization, or equivalent secure methods such as JWT-based authentication and tokenized access control for services.
- Enforces Role-Based Access Control (RBAC) wherever applicable to ensure least-privilege access and secure authorization across services.

4.2 Data Protection & Encryption

- All sensitive data, including token keys, is protected using AES-256 encryption, and passwords are securely hashed using industry-standard cryptographic algorithms to ensure data security and compliance.
- Encryption is enforced for data in transit using TLS 1.2 or higher

4.3 Audit & Monitoring

- Comprehensive audit trails are implemented to ensure full traceability of user activities, capturing who performed specific actions, when, and within which context.
- Centralized logging and monitoring mechanisms are implemented for real-time tracking and alerting of security events

4.4 Application Security

- Comprehensive input validation and sanitization controls are enforced to prevent common web vulnerabilities, including SQL injection and cross-site scripting (XSS), across all applicable interfaces.

4.5 DevSecOps & CI/CD Security

- Implements a DevSecOps pipeline to ensure security is integrated throughout the CI/CD process.

4.6 Database & Infrastructure Security

- Production database access is strictly restricted to authorized application services, with all external access disabled to ensure maximum data security and compliance.
- External system secrets and keys are securely stored, managed, and access-controlled to ensure they remain inaccessible to unauthorized systems, services, or users.

4.7 Data Exchange & Interoperability

- Ensures all data exchange formats conform to globally recognized standards, such as JSON and/or XML, to maintain interoperability, consistency, and compliance across systems.
- Ensures full compliance with GDPR (General Data Protection Regulation), the European Union's data privacy law that governs the lawful collection, processing, storage, and protection of personal data, including user rights and data security requirements.
- Architected to support the implementation of SOC 2 standards in subsequent phases, ensuring readiness for future compliance and audit requirements.
- Ensures that unique identifiers are consistently preserved, securely managed, and protected during integration with Techconnectr 2.0 to maintain data integrity, traceability, and security.

5 High-Level Workflow

1. **Ingestion:** User uploads a CSV/JSON of leads or contractors. If required, an integration with source system can be implemented on need basis.
2. **Parallel Processing:** The system triggers multiple API calls (ZeroBounce, Google, etc.) and web scrapers simultaneously, rate limit may be applied according to the external APIs limitations.
3. **AI Voice Trigger:** The bot initiates call for phone verification in the background.
4. **Consolidation:** The engine aggregates data and calculates the confidence score.
5. **Human Review:** A clean UI displays "Needs Attention" items for the user to manually validate.
6. **Export:** Final validated list is generated.

6 Confidence Score Logic (The "Brain")

To reach that 3-hour window, the system needs to make its own decisions. We use a weighted scoring model where different sources have different "trust" levels.

6.1 The Scoring Formula

The system calculates a Final Confidence Score (SS_{total}) for each lead. If $SS_{total} \geq 85\%$, it's auto-passed. If $SS_{total} < 85\%$, it hits the manual queue.

Data Point	Weight (W)	Verification Criteria
Email	30%	ZeroBounce "Valid" status + TowerData match.
Identity	25%	LinkedIn Job Title or Company Website title matches the entered data. In case if the data not available in LinkedIn or company website, application should check in other professional sources like google, Glassdoor etc. these details should be embedded in application configuration.
Phone	20%	AI Bot confirms the person is reachable/still employed.
Company	15%	LinkedIn industry/size matches Glassdoor/Bloomberg.
Location	10%	Google Maps API returns a "Partial Match" or "Exact Match".

Example Logic:

If an email is "Catch-all" (risky) but the LinkedIn title and Phone Bot both confirm the person is active, the score might hit 80%—flagging it for a quick 10-second human "sanity check" rather than a full 20-minute manual search.

7 AI Voice Bot Workflow

This replaces the most time-consuming manual task: cold-calling to verify a contractor or lead.

7.1 The Workflow: "The Verification Ping"

1. Trigger: The system pulls the phone number from Sales Navigator/LinkedIn.
2. Initiation: The AI Bot (using a tool like Bland AI or Vapi) places a call during local business hours.
3. The Script (Sample):
"Hi, I'm an automated assistant calling to verify contact details for [Name]. Is [Name] still the [Job Title] at [Company Name]?"

4. Speech-to-Text (STT) Analysis:
 - "Yes": AI logs "Confirmed" (Score: 100%).
 - "No longer work here", "No": AI logs "Invalid" (Score: 0%).
 - Voicemail/No Answer: AI schedules one retries in 60 minutes.
5. If the lead is a tele marketing lead, the full call will be recorded and transcript and both record and transcript text will be attached to the respective record.
6. Output: The transcript and a sentiment tag (Success/Fail/Uncertain) are attached to the lead record.

8 Assumptions & Limitations

1. Queue Manager mechanism will be implemented to trigger parallel verifications to optimize the time required to complete the verifications.
2. The number of parallel verifications should be configured based on the target website rate limits (ex: linked in)
3. The verification time will vary depending on size of the lead content.
4. The infrastructure is estimated to the minimum requirements to optimize the initial cost. It can be scaled according to the requirements.
5. Implementation efforts will be in Indian Rupees.
6. The subscription cost is provided in both Indian Rupees and source currency (like US dollars) for the each of understanding.
7. Subscription costs are indicative only. They may vary. Client needs to pay the actuals to the respective company.
8. If developer need to provide the all the subscriptions, we will add 10% on top of the actuals as administrative cost for procuring the subscriptions.

9 Hosting Environment

Below are minimum recommended hosting specifications (sized for ~100,000 leads workload) and indicative monthly costing for Azure, AWS, and On-Premises options. All the costs are estimated considering the cloud will be hosted in India regions of respective cloud providers. These costs are indicative only; actuals will be calculated during the deployment based on the sizing requirements.

9.1 Azure Hosting Option (Recommended)

9.1.1 Application & Processing Layer

Component	Azure Service	Specification
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Backend APIs (Fast API/Django)	Azure App Service / AKS	2 instances, 4 vCPU, 16 GB RAM each
Worker Nodes (Queue Processing)	VM Scale Set / AKS	2 instances, 8 vCPU, 32 GB RAM
Scheduler/Jobs	Azure Functions	Consumption / Premium

9.1.2 Database & Cache

Component	Azure Service	Specification
Database	Azure Database for PostgreSQL	4 vCore, 128GB storage and scalable
Cache	Azure Cache for Redis	Standard C2 / C3

9.1.3 Storage & Logs

Component	Azure Service	Specification
File Storage (Call recordings/transcripts, exports)	Azure Blob Storage	500GB–2TB
Logging & Monitoring	Azure Monitor + Log Analytics	Centralized logging

9.1.4 Security

Component	Azure Service
Secrets	Azure Key Vault
Network Security	NSG + Private Endpoints
WAF	Azure Application Gateway WAF

9.2 AWS Hosting Option (Recommended)

9.2.1 Application & Processing Layer

Component	AWS Service	Specification
Backend APIs	ECS Fargate / EKS	2 tasks, 4 vCPU, 16GB RAM
Worker Nodes	ECS/EKS or EC2 Auto Scale	2 instances, 8 vCPU, 32GB RAM
Scheduler	Lambda / EventBridge	Scheduled jobs

9.2.2 Database & Cache

Component	AWS Service	Specification
Database	RDS PostgreSQL	db.m6i.xlarge (4 vCPU, 16GB) and scalable
Cache	ElastiCache Redis	cache.m6g.large / scalable

9.2.3 Storage & Logs

Component	AWS Service
Storage	S3
Logs	CloudWatch
Monitoring	CloudWatch Alarms + X-Ray

9.2.4 Security

Component	AWS Service
Secrets	AWS Secrets Manager
WAF	AWS WAF + ALB
Networking	VPC + Security Groups

9.3 On-Premises Hosting Option (Minimum Recommended)

9.3.1 Minimum Infrastructure (100,000 Leads)

Component	Specification
Application Server Nodes	2 servers (16 cores CPU, 64GB RAM each)
Database Server	1 server (16 cores CPU, 128GB RAM, RAID SSD storage)
Storage/NAS	2TB SSD + 4TB HDD (for call recordings/logs/exports)
Backup Server	1 server (8 cores, 32GB RAM, 4TB storage)
Firewall + Load Balancer	Fortinet / Sophos / equivalent
Network	1Gbps LAN, secure DMZ

9.3.2 Software Stack (Typical)

- Ubuntu / RHEL
- Docker + Kubernetes (optional)
- PostgreSQL Cluster (Primary + Replica recommended)
- Redis
- Prometheus + Grafana (monitoring)
- ELK Stack (logging)

10 Subscriptions:

Service	Starting Price (10k Emails)		Model	Key Advantage
	USD	INR		
ZeroBounce	\$75	~ ₹ 7,083.00	PAYG	10,000 credits
Bouncer	~\$60 - \$80	~ ₹ 5,666.40 - ₹ 7,555.20	PAYG / Sub	High accuracy and easy-to-use API.

NeverBounce	~\$80	~ ₹ 7,555.20	PAYG	Industry standard for "Sync" cleaning workflows.
MyEmailVerifier	\$25	~ ₹ 2,361.00	PAYG	Extremely affordable (\$0.0025/credit).
MillionVerifier	~\$6	~ ₹ 566.64	PAYG	Best for high-volume, low-cost bulk cleaning.
Emailable	~\$30 - \$50	~ ₹ 2,833.20 - ₹ 4,722.00	Sub / PAYG	Modern UI and excellent Zapier/API integrations.
Verifalia	~\$49	~ ₹ 4,627.56	Sub / PAYG	Flexible plans with 25 daily free credits.
Deliverability Suite	~\$49	~ ₹ 4,627.56	monthly	To ensure the email domain is trusted by ISP providers like google and looks and monitor block listing

Need	Recommended Tool	Estimated Entry Cost	
		USD	INR
Phone Status	Twilio Lookup	\$0.01 / request	₹ 0.94 / request
Identity/LinkedIn	Apollo.io	\$49 / month	₹ 4,627.56 / month
Company Details	Clearbit / Clay	\$0.15 / company	₹ 14.17 / company
Address/Maps	Google Places	\$0.002 / request	₹ 0.19 / request
Voice AI Agent	Vapi	\$0.05 / minute	₹ 4.72 / minute

11 Project Schedule

11.1 Project deliverables by phase

11.1.1 Phase 1: Planning

- Kickoff
- Requirements finalization
- Project plan & milestones

11.1.2 Phase 2: Design

- Architecture & database design
- UI wireframes
- Security & compliance design

11.1.3 Phase 3: Core Development

- User login + RBAC
- Lead upload module
- Admin configuration module

11.1.4 Phase 4: Verification Engine

- Email, phone, address, website verification modules
- Parallel processing/queue framework

11.1.5 Phase 5: AI & Scoring

- AI voice bot integration
- Confidence score logic
- Manual review workflow

11.1.6 Phase 6: Dashboard & Reports

- Verification dashboard
- Human review screens
- Export reports (CSV/Excel)

11.1.7 Phase 7: Testing

- SIT testing
- Performance testing
- Security validation

11.1.8 Phase 8: Deployment

- UAT support
- Production deployment
- Go-live & handover

11.2 Project Milestones

SI.NO	Milestone	Target Week
1	Requirements Signoff	?
2	Architecture Signoff	?
3	Sprint-1 Demo	?
4	Verification Engine Ready	?
5	Voice Bot Demo	?
6	MVP Feature Complete	?
7	SIT + Performance Signoff	?
8	UAT Signoff + Go-Live	?

11.3 Project Schedule

1 to ? weeks etc. This is just indicative.

12 Phased approach for implementation

Based on the request to split the implementation in 2 phase / deliveries, below is the split up of implementation along with the costing split.

12.1 Phase 1 – Core Verification Platform

Goal: Deliver a usable production system for lead verification.

12.1.1 Functionalities

12.1.1.1 Planning & Design

- Requirements finalization
- Architecture design
- Database design
- UI wireframes
- Security design

12.1.1.2 Platform Foundation

- User Login
- RBAC
- User Management
- Admin Configuration Module

12.1.1.3 Lead Processing

- CSV/JSON Upload
- Lead Processing Framework
- Queue Manager

12.1.1.4 Verification Engine (Core)

- Email Verification
 - SMTP Verification
 - External API Integration Framework
 - ZeroBounce Integration

- TowerData / NeverBounce configuration support
- Website Verification
 - DNS Validation
 - URL Validation
- Address Verification
 - Google Maps API integration

12.1.1.5 Basic Dashboard

- Upload Status
- Verification Status
- Processing Statistics

12.1.1.6 Testing

- Unit Testing
- SIT for Core Modules
- Security Validation for Core Platform
- UAT for Phase 1 Features

12.1.1.7 Deployment

- Phase 1 Production Deployment
- User Training
- Go-Live Support

12.1.1.8 Phase 1 Deliverable

Operational Verification Platform with Email, Website and Address Validation

12.2 Phase 2 – AI Verification & Decision Engine

12.2.1 Functionalities

12.2.1.1 Advanced Verification

- Phone Verification using AI Voice Bot
- LinkedIn/Sales Navigator integration
- Job Title Verification
- Company Size Verification
- Industry Verification
- Company Website Cross Verification

12.2.1.2 AI Voice Bot Workflow

- Vapi / Bland AI Integration
- Call Recording
- Transcript Generation
- Verification Ping Workflow
- Sentiment Classification
- Retry Logic

12.2.1.3 Confidence Score Engine

- Weighted Scoring Logic
- Auto Pass
- Manual Queue
- Fail Classification

12.2.1.4 Human Review Workflow

- Manual Validation Queue
- Comparison Screens
- Exception Handling

12.2.1.5 Dashboard & Reporting

- Verification Dashboard
- Human Review Screens
- Export Reports (CSV/Excel)

12.2.1.6 Testing

- SIT for AI Components
- Performance Testing
- End-to-End Testing
- Security Validation
- Final UAT

12.2.1.7 Deployment

- Production Upgrade
- User Training
- Go-Live
- Handover

12.2.1.8 Phase 2 Deliverable

Complete Lead Verification System